天津大学 Tianjin University

TANKLab Tianjin Key Laboratory of Advanced Networking

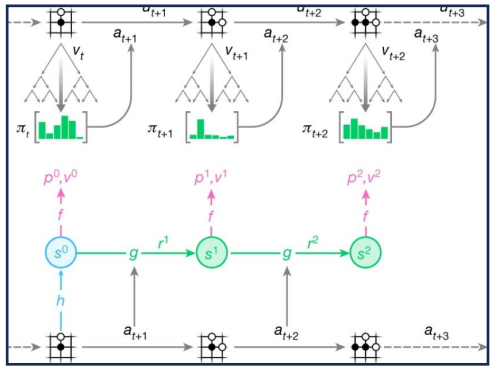
High-throughput Sampling, Communicating and Training for Reinforcement Learning Systems

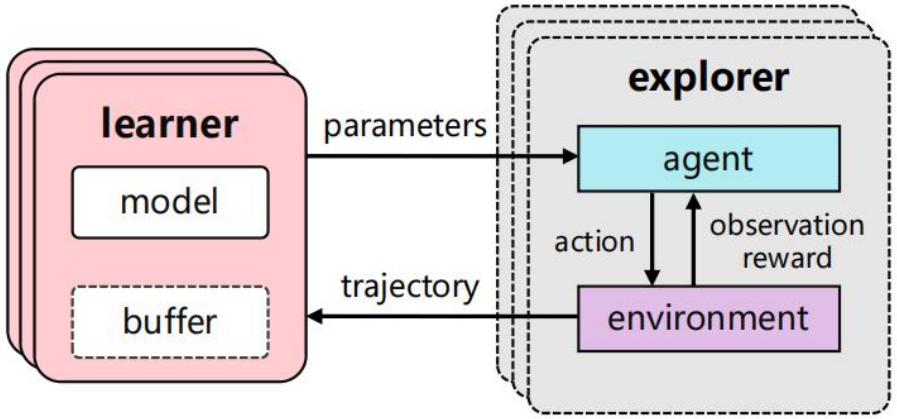
Laiping Zhao[1], Xinan Dai[1], Zhixin Zhao[1], Yusong Xin[1], Yitao Hu[1],
Jun Qian[2], Jun Yao[2], and Keqiu Li[1]

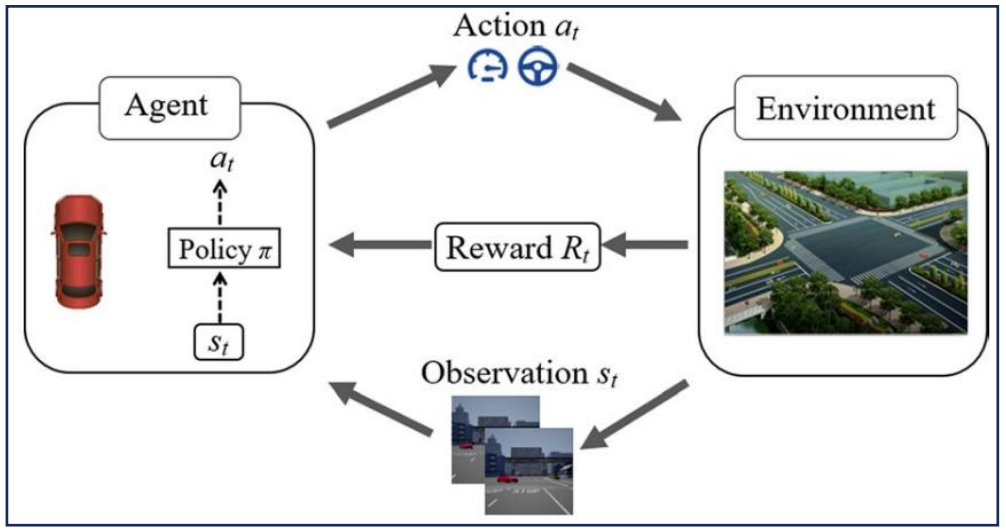
Tianjin University[1], HUAWEI[2]

Background

- RL’ s applications: Go chess, autonomous driving, robot manipulation and others.
- RL’ s limits: low sample efficiency, complex implementation, slow training speed.

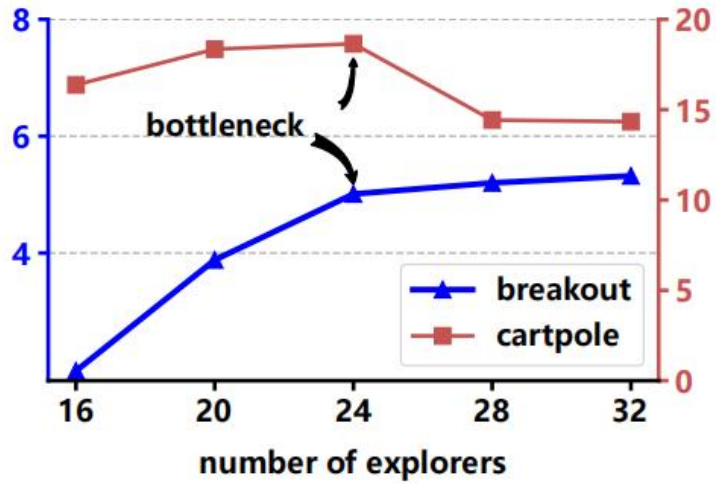




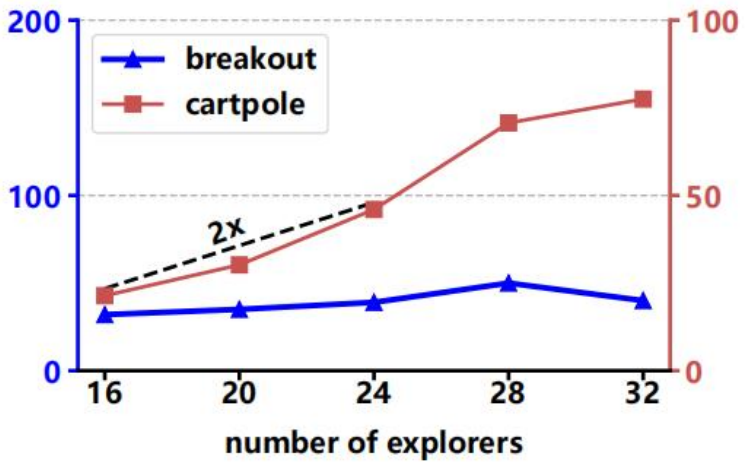


Motivation

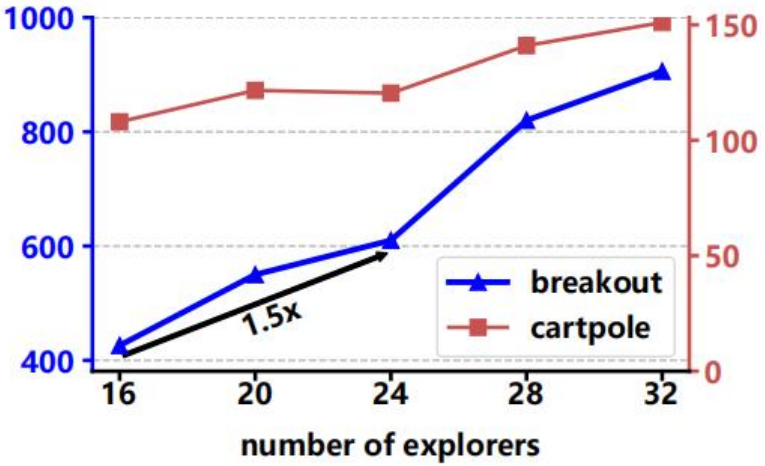
- **The bottleneck in the throughput of the same model may vary over the simulation environments.**



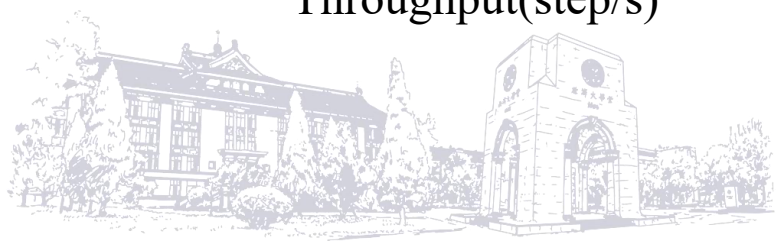
Throughput(step/s)



Train time(ms)

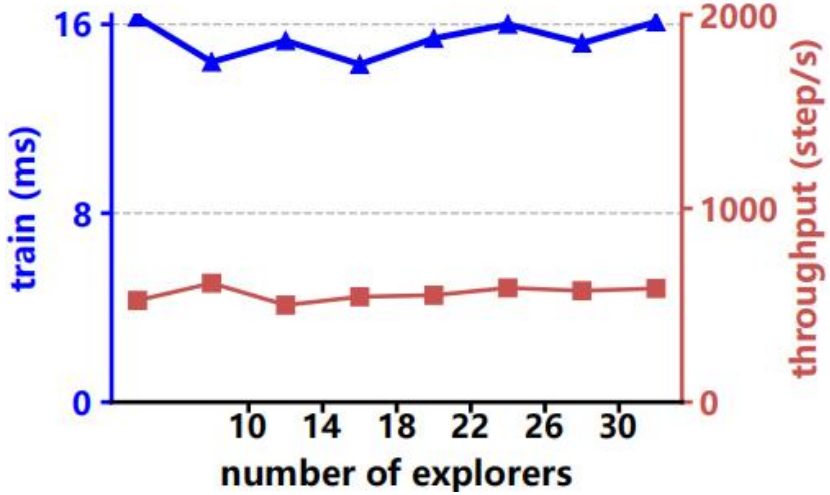


Sampling time(ms)

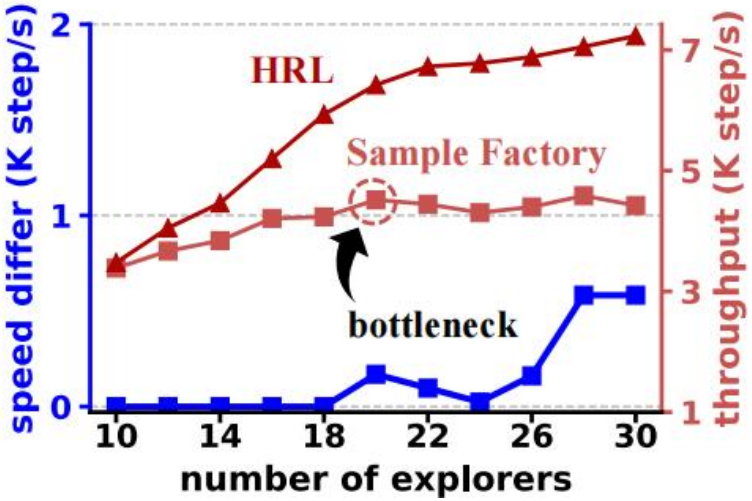


Motivation

- Under the network bottleneck, the throughput cannot be improved by optimizing the explorers.



- Due to the uncertain location of bottlenecks, a single optimization technique cannot guarantee an overall improvement in throughput.



Motivation

- **Bottlenecks may occur in many places. A holistic optimization method that can improve the overall throughput of RL systems is needed.**

- **Challenges:**

1) **Discover and identify the location of bottlenecks**

2) **Effective optimization techniques**

3) **Balance the cost and bonus of different techniques and performance of different modules**

Design

➢ Architecture

- **Coordinator:**

  Discover bottlenecks' locations and schedule different modules

- **Pipeline sampler:**

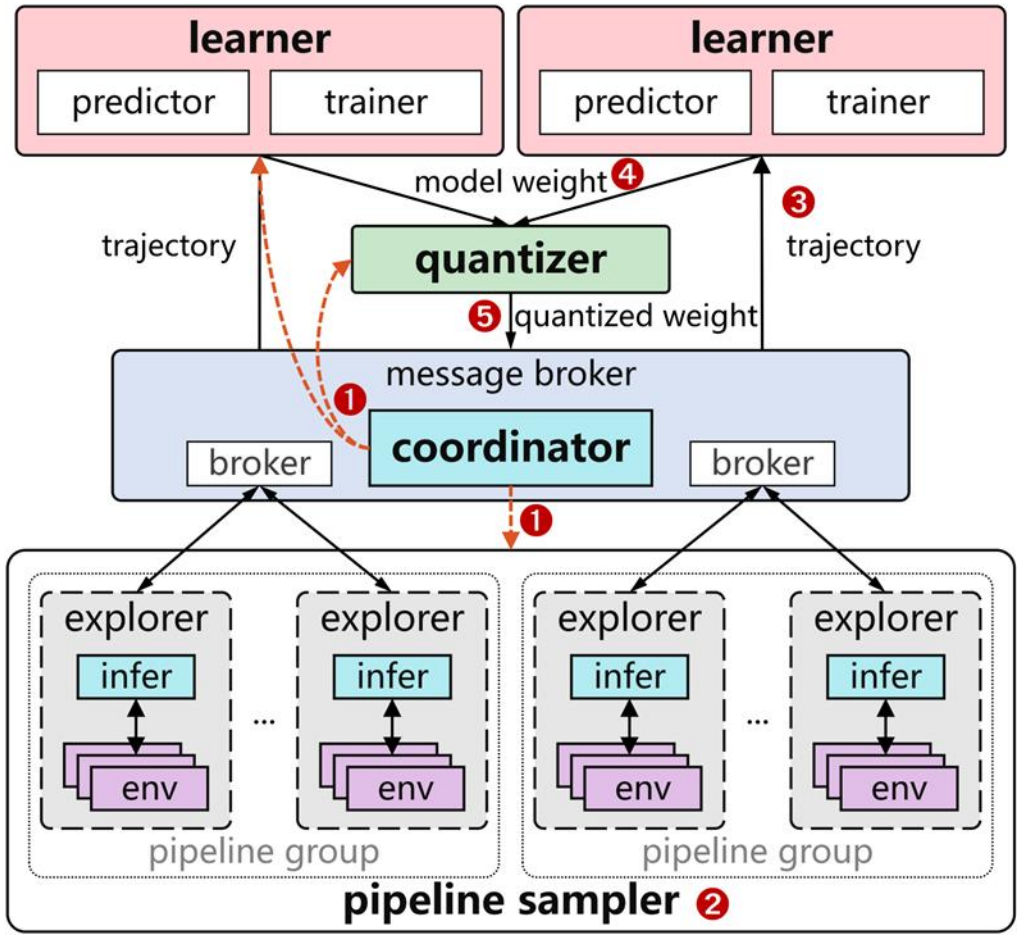
  Use groups of inference and simulation pipelines to speed up sampling tasks.

- **Multi-Learner:**

  Distribute the training workload across multiple GPUs.

- **Communication:**

  Use buffer and quantization to mitigate the network bottleneck.

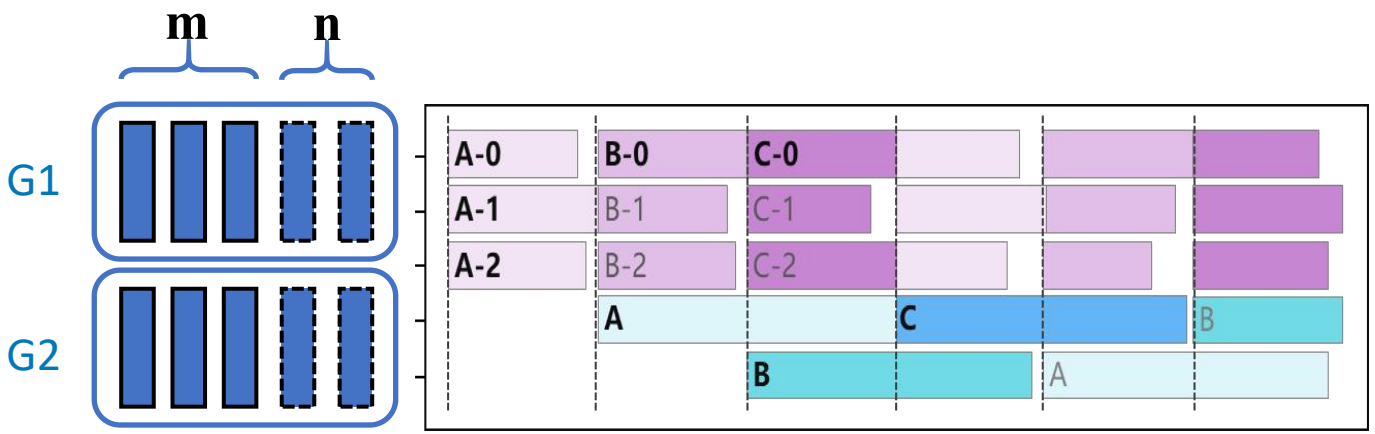


Design

➢ Pipeline Sampler

- **Resource Allocation:**

  A group occupies m+n CPU cores

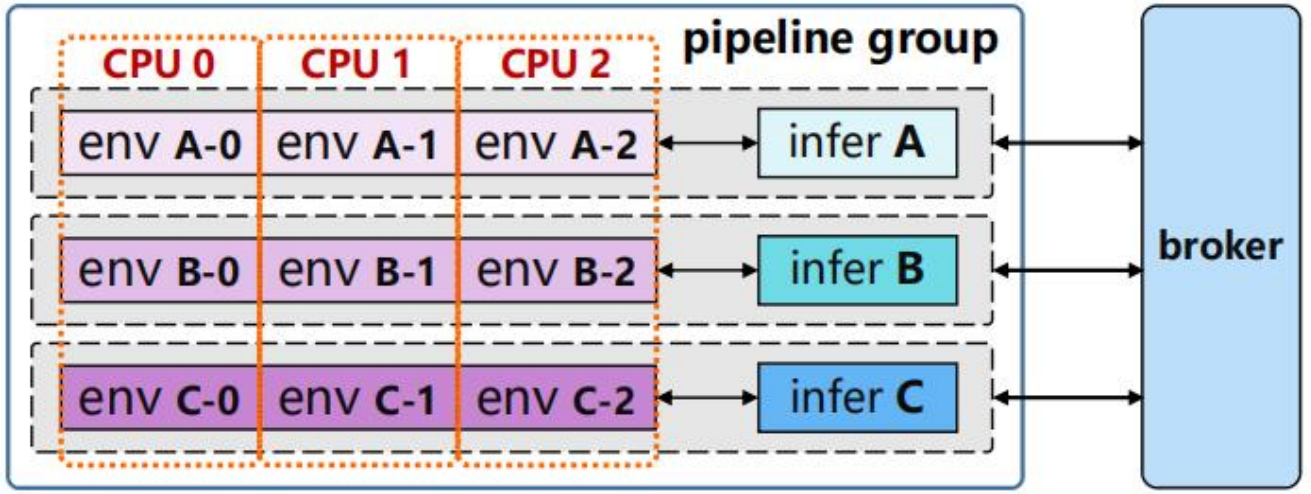


- **Timeline:**

  (1) The environment processes in EA are executed

  (2) The data is handed over to the inference process

  (3) Release the env processes in EA, and start to execute the env processes in EB.



Evaluation

➢ Experimental Setup

- **RL Benchmarks:**

  DQN, PPO, and IMPALA

- **Performance Metrics:**

  (1) Throughput (step per second)
  (2) Average episode return (or reward/return)

- **Competing Systems:**

  Sample Factory, Rllib, and XingTian

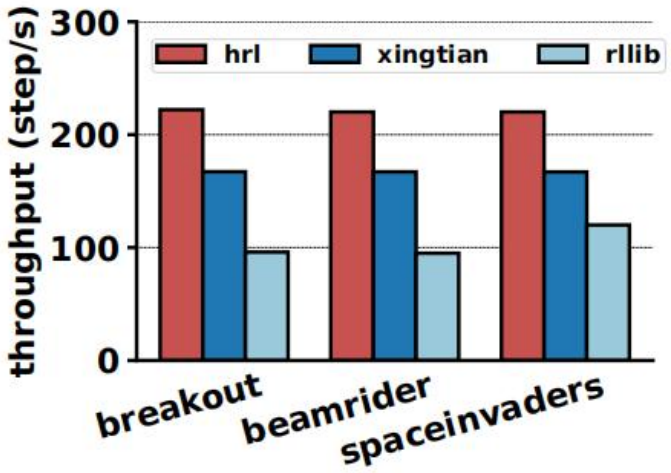
Evaluation

- **HRL increases the throughput of RL system in different scenes with unchanged convergence:**

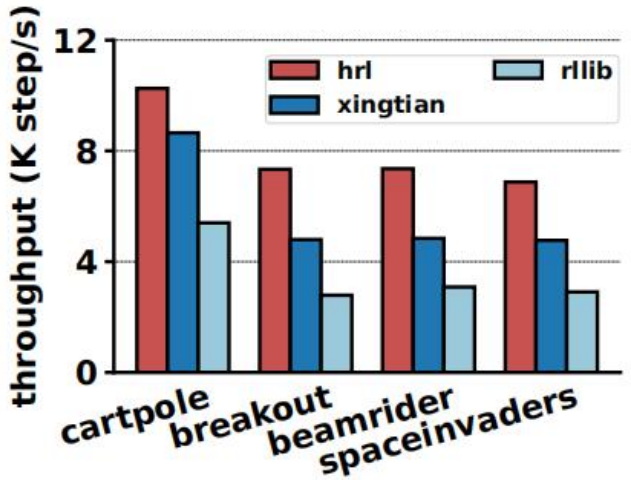
  (1) HRL brings 31.7% – 32.8% increase in DQN’s throughput

  (2) HRL brings 18.7%–90.6% increase in PPO’s throughput

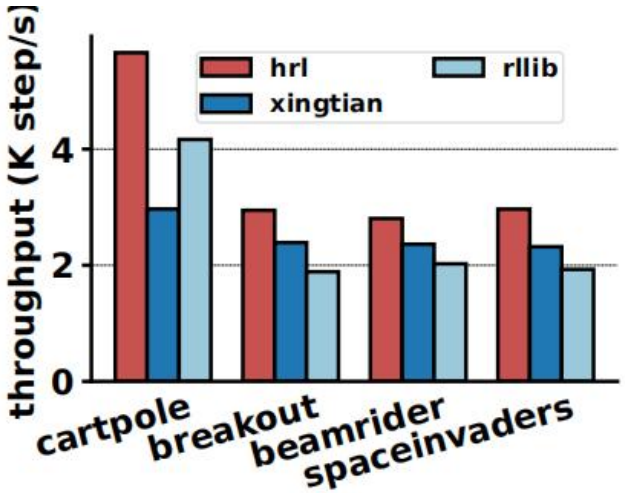
  (3) HRL brings 18.6% – 52.9% increase in IMPALA’s throughput



(a) DQN



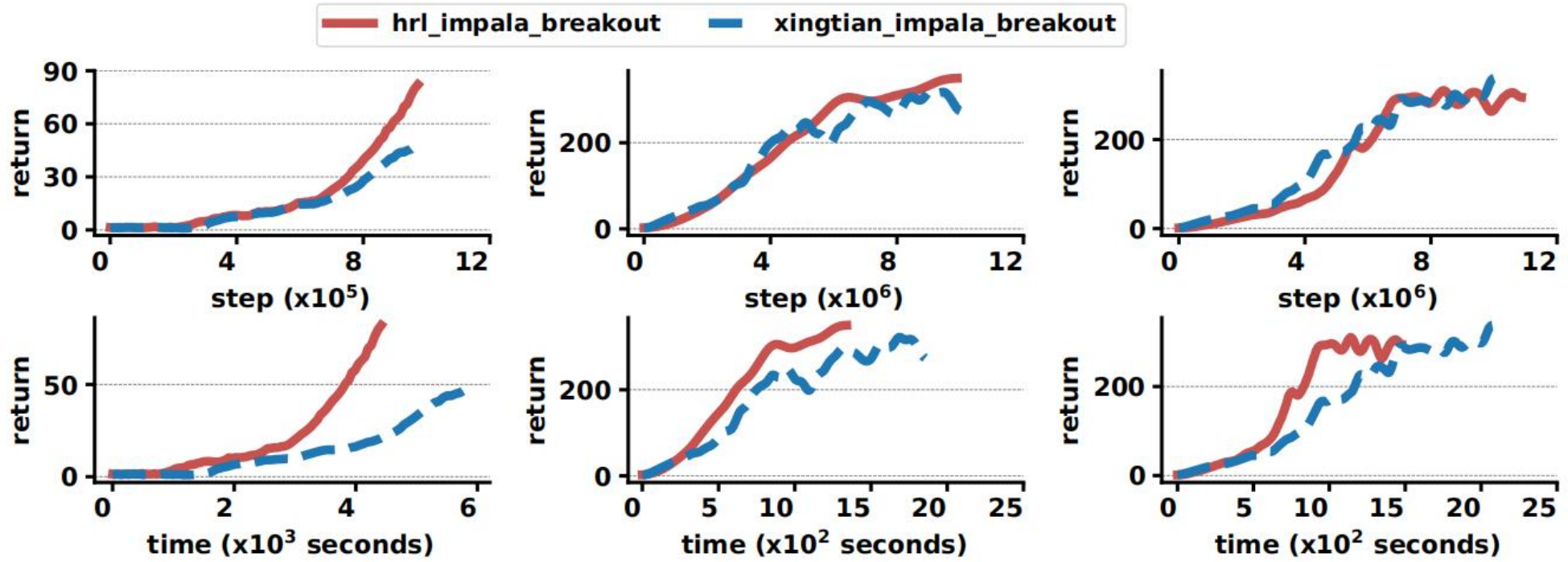
(b) IMPALA



(c) PPO

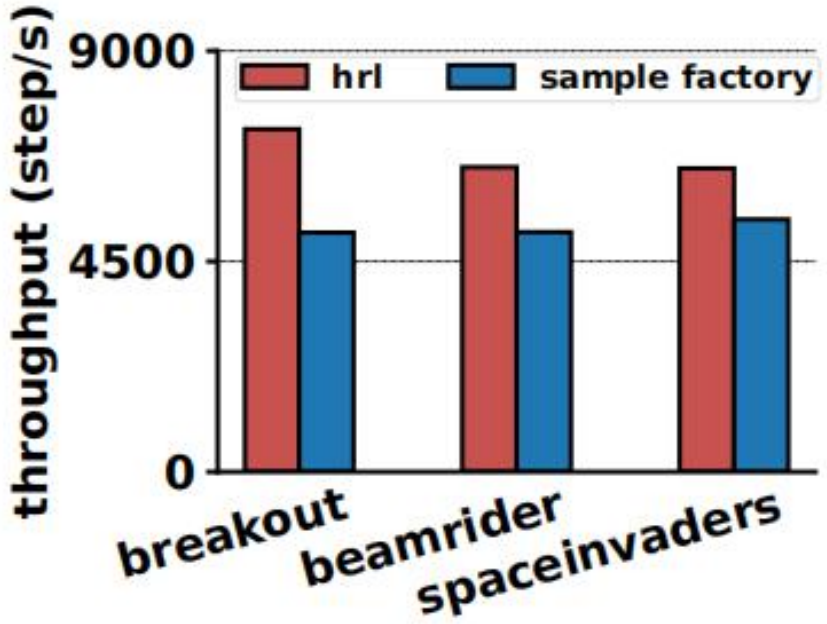
Evaluation

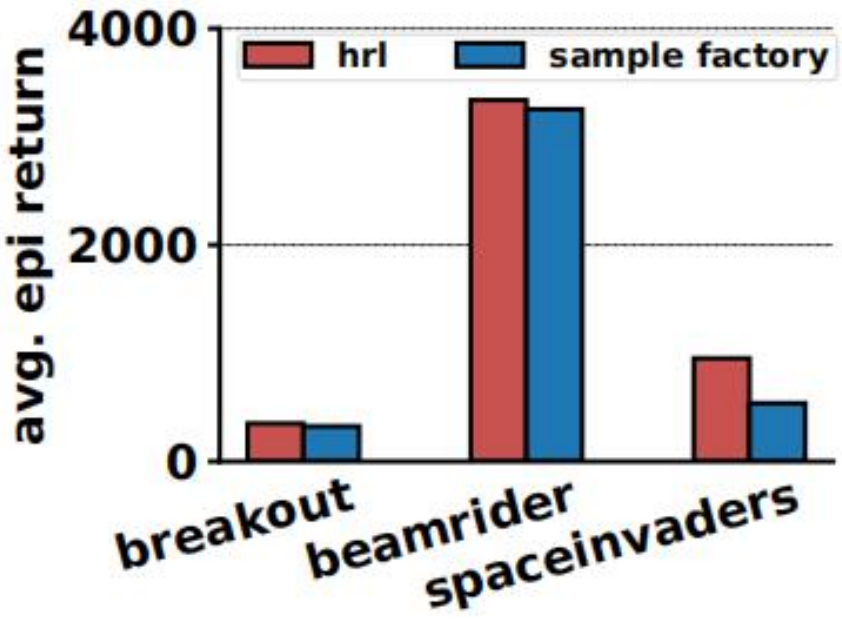
- **HRL has a faster convergent speed than XingTian, with a speed increase of 36% to 41%**



Evaluation

- **HRL outperforms the local optimization system of Sample Factory in the overall throughput.**
- **Compared with Sample Factory, HRL improves the average episode return by 8.0%-77.1%.**





Conclusion

1. We provide **a holistic analysis of the bottlenecks** in the RL training.
2. We have proposed a **group-paralleled pipeline** sampler and applied the **data quantization** and **multi-learner training** into the system.
3. We design a **coordination method** to make the above modules work together.
4. The experiment results show that it can improve the DQN throughput by 31.7% – 32.8% on average, PPO by 18.7 – 90.6%, and IMPALA by 18.6% – 52.9%.

Thank you